



CHAPTER

9

Mechanical Properties of Solids

Sr. No.	Concept	Formulas	Other information
1.	Stress	$\text{Stress} = \frac{\text{Restoring force}}{\text{Area}} = \frac{F}{A}$	
2.	Strain	Longitudinal strain = $\frac{\Delta l}{l}$ Volumetric strain = $\frac{\Delta V}{V}$ Shear strain:  (Here ϕ is very small) $\tan \phi = \frac{x}{L}, \phi = \frac{x}{L}$	Δl = change in length. l = original length. ΔV = change in volume. V = original volume. ϕ = shear strain or angle of shear.
3.	Hooke's law	Stress $\propto$ Strain or Stress = $E \times$ strain	E = modulus of elasticity of the material
4.	Young's modulus of elasticity (Y)	$Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{\frac{F}{A}}{\frac{\Delta l}{l}} = \frac{F}{A} \cdot \frac{l}{\Delta l}$	Δl = change in length l = original length
5.	Bulk modulus of elasticity (k)	$k = \frac{\text{Normal stress}}{\text{Volumetric strain}} = -\frac{\frac{F}{A}}{\frac{\Delta V}{V}}$ $= -\frac{p}{\frac{\Delta V}{V}} = -\frac{pV}{\Delta V}$	p = pressure. ΔV = change in volume. V = original volume.
6.	Compressibility (C)	$C = \frac{1}{\text{Bulk modulus}} = \frac{1}{k}$	k = bulk modulus of elasticity.

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7.	Modulus of rigidity	$\eta = \frac{\text{shearing stress}}{\text{shearing strain}} = \left(\frac{\frac{F_{\text{tangential}}}{A}}{\phi} \right) = \frac{F_{\text{tangential}}}{A\phi}$	$F_{\text{tangential}}$ = tangential force. ϕ = shear strain. A = area.
8.	Poisson's ratio	Poisson's ratio = $\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$ Or $\sigma = \frac{\frac{\Delta D}{D}}{\frac{\Delta l}{l}}$	σ = poisson's ratio. ΔD = change in diameter. D = original diameter.
9.	Relations between Y, k, η and σ	(i) $Y = 3k(1 - 2\sigma)$ (ii) $Y = 2\eta(1 + \sigma)$ (iii) $\sigma = \frac{3k - 2\eta}{6k + 2\eta}$ (iv) $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{k}$	Y = Young's modulus of elasticity. k = Bulk modulus of elasticity. η = modulus of rigidity. σ = poisson's ratio.
10.	Elastic potential energy	Total P.E. stored in a stretched wire, $U = \frac{1}{2} \times \text{Stretching force} \times \text{extension} = \frac{1}{2} F\Delta l$. or $U = \frac{1}{2} (\text{Stress} \times \text{strain} \times \text{volume of wire})$ P.E. stored per unit volume of a stretched wire, $u = \frac{1}{2} (\text{Stress} \times \text{strain})$ or $u = \frac{1}{2} (\text{Young's modulus} \times \text{strain}^2)$	F = applied force. Δl = change in length. u = energy per unit volume.